Raul Steleac

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Professional Experience

Research Intern May 2023 – Dec. 2024
Dematic *Remote*

- Researched **hierarchical multi-agent reinforcement learning** methods in a specialised, realistic warehouse simulation environment in collaboration with domain experts.
- Developed a scalable architecture that outperformed industry-standard heuristic approaches in deliveries per hour, **demonstrating improved operational performance**. Published at **IROS 2024** [2].
- Built **TA-RWARE** [4], an open-source simulator for studying the **multi-agent task assignment** problem.

Machine Learning Scientist Jan. 2023 – Aug. 2023
SenseStreet *London, United Kingdom*

- Developed transformer-based architectures for a **natural language processing** pipeline to extract financial information from anonymised investment banking conversations, aimed at supporting traders in making **more efficient and precise deals**.

Machine Learning Engineer Nov. 2021 – Jan. 2023
Healx *Cambridge, United Kingdom*

- Developed **natural language processing** methods to mine large-scale biomedical datasets, enabling the construction of knowledge graphs to support scientists in **rare disease drug discovery**.
- Designed and implemented a transformer-based contextual entity linking architecture to disambiguate and map in-sentence entities to biomedical ontologies.
- Constructed a noise-aware training approach that utilises ensemble models to build loss-masks for enhanced named entity recognition training, **increasing the F1 score of the production model by 7.5%**.

Software Development Intern Dec. 2018 – Aug. 2020
Intel Software Development *Timisoara, Romania*

- Contributed to the development of two generations of the Intel Movidius Visual Processing Unit, enabling **accelerated neural network inference** for real-time applications including drones and robots.

Junior Software Developer Oct. 2017 – Dec. 2018
Nokia Networks *Timisoara, Romania*

- Diagnosed and resolved software issues in C++ using object-oriented methodologies, delivering rapid fixes within the Fault Detection and Alarm Raising department.

Education

University of Edinburgh Edinburgh, United Kingdom
PhD in Robotics and Autonomous Systems *Sept. 2023 – Present*

- **Supervisors**: Mohan Sridharan and Amos Storkey.
- **Topic**: Discovery and Reuse of Coordinated Behaviours in Multi-Agent Reinforcement Learning.
- **Key Areas**: Multi-Agent Systems, Reinforcement Learning, Temporally Extended Actions, Options Framework, Hierarchical Reinforcement Learning and Robotics.

Imperial College London London, United Kingdom
MSc in Computing (Artificial Intelligence and Machine Learning) *Oct. 2020 – Oct. 2021*

- **Award**: Distinction.
- **Studied modules**: Reinforcement Learning, Deep Learning, Natural Language Processing, Computer Vision, Machine Learning for Imaging, Probabilistic Inference.

Polytechnic University Timisoara — Timisoara, Romania
BEng in Computers and Information Technology — *Sept. 2016 – July 2020*

- **Grades**: 10.0/10.0 thesis final grade and 9.20/10.0 module average grade.
- Received **merit scholarships** in 7 out of the 8 semesters.

Publications

Conferences and Journals

[1] **Raul D. Steleac**, Mohan Sridharan and David Abel; "Inter-Agent Relative Representations for Multi-Agent Option Discovery", in **ICLR** 2026.

[2] A. Krnjaic, **Raul D. Steleac**, J. D. Thomas, G. Papoudakis, L. Schäfer, A. W. Keung To, K. Lao, M. Cubuktepe, M. Haley, P. Börsting and S. V. Albrecht; "Scalable Multi-Agent Reinforcement Learning for Warehouse Logistics with Robotic and Human Co-Workers", in **IROS** 2024.

Theses

- **Raul D. Steleac**, "Curriculum Reinforcement Learning: Re-visiting Tabular Methods", Master's Thesis, Imperial College London, 2021.
- **Raul D. Steleac**, "End-to-end Speech Emotion Recognition", Bachelor's Thesis, Polytechnic University Timisoara, 2020.

Open-Source Software Contributions

[3] **IARO**: reference implementation of **inter-agent relative representations** [1], with a JAX-based framework for eigenoption discovery and option reuse in multi-agent systems.

[4] **TA-RWARE**: warehouse-inspired simulator for studying **multi-agent task assignment**, released with [2] and subsequently adopted as a benchmark in multi-agent reinforcement learning research.

Academic Engagement

- **Reviewer**, International Conference on Learning Representations (**ICLR** 2026).
- **Reviewer**, Journal of Machine Learning Research (**JMLR**).
- **Reviewer**, International Conference on Intelligent Robots and Systems (**IROS** 2024).

Awards

- **UKRI Studentship**: Funded place on the EPSRC **Robotics and Autonomous Systems** Centre for Doctoral Training programme.
- **InGeniously**: Won **first place** in an undergraduate competition for developing a self-driving robot car that autonomously navigated miniature mazes via wirelessly transmitted routes while following traffic rules.
- **Zero Robotics**: **Semifinalist** in a NASA-organised international high-school programming competition, developing algorithms for SPHERES robots aboard the ISS to autonomously photograph meteorite subjects.

Technical Skills

Programming languages:	Python, SQL, C++, C, Java, Bash
Frameworks:	JAX, PyTorch, HuggingFace, TensorFlow, Keras
Developer Tools:	Git, AWS, Docker, CircleCI, Jira, Google Cloud Platform, BigQuery, Airflow
Libraries:	NumPy, Pandas, Pydantic, pytest, MuJoCo, Matplotlib, Librosa